

Differential Spectral Damping: Gap-Adaptive Regularization for Ill-Conditioned Kernel Methods

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Abstract—Kernel methods requiring matrix inversion—particularly Least-Squares Twin SVMs—suffer from exponential eigenvalue decay in their system matrices, producing severely ill-conditioned problems where standard Tikhonov regularization applies uniform damping regardless of eigenvector reliability. We propose Differential Spectral Damping (DSD), a regularization formula that adapts its penalty to localized eigengap structure: preserving eigenvectors with large spectral gaps (reliable per Davis-Kahan theory) while suppressing those with small gaps (directionally corrupted). Through rigorous paired testing with fairly optimized baselines (including gradient-optimized Tikhonov), we demonstrate that DSD improves LSTSVM classification accuracy by +4.8pp on real-world GINA ($d = 970$, Cohen’s $d = 4.49$), +10.4pp at $d = 200$, and +2.6pp on Madelon ($d = 500$, Cohen’s $d = 1.76$)—all over optimally-tuned Tikhonov ($p < 0.0001$) using only principled spectral initialization while Tikhonov receives grid search. For pre-image reconstruction, DSD matches Tikhonov ($p = 0.99$); both reduce naive error by $66\times$. We characterize the precise operating regime and document where simpler methods suffice.

Index Terms—Kernel methods, spectral regularization, eigengap adaptation, support vector machines, ill-conditioned systems, Twin SVM, numerical stability.

I. INTRODUCTION

Kernel methods map data into high-dimensional feature spaces via the kernel trick, enabling non-linear classification without explicit feature computation [1], [2]. However, methods requiring inversion of cross-class product matrices—particularly Least-Squares Twin SVM (LSTSVM) [3]—face severe ill-conditioning: these product matrices ($E_1^\top E_1 + c^{-1} E_2^\top E_2$) empirically exhibit extreme spectral tail-clustering, where at $d = 200$ the bottom 50% of eigenvalues occupy only 7% of the total spectral range.

LSTSVM remains actively deployed in domains where deep learning overfits due to limited samples [4]: genomic classification ($d = 1000$ – 20000 , $n = 50$ – 500), spectroscopic analysis, and regulatory credit scoring with engineered features. In these settings, a +3–10 percentage point accuracy improvement from regularization alone directly impacts downstream decision quality.

The Davis-Kahan $\sin(\Theta)$ theorem [5] establishes that eigenvector perturbation under noise is bounded by $\|E\|/\delta_i$, where δ_i is the eigengap. Tightly-clustered eigenvalues correspond

to eigenvectors that are directionally corrupted—regardless of whether eigenvalues are subsequently regularized. Tikhonov regularization $(W + \gamma I)^{-1}$ addresses magnitudes but ignores this directional unreliability: a single scalar γ cannot simultaneously avoid over-regularizing the well-separated head and under-regularizing the clustered tail.

We propose **Differential Spectral Damping (DSD)**, which adapts its penalty to the localized eigengap density:

$$\tilde{\lambda}_i^{-1} = \frac{\lambda_i}{\lambda_i^2 + \alpha \cdot \exp(-\beta \cdot \delta_i)} \quad (1)$$

The exponential form is a design choice selected for smoothness, differentiability, and natural saturation behavior; alternative functional forms (sigmoid, polynomial) would provide similar qualitative behavior. The localized bilateral eigengap $\delta_i = \min(|\lambda_i - \lambda_{i-1}|, |\lambda_i - \lambda_{i+1}|)$ approximates the full spectral separation Davis-Kahan uses, exact for monotonically decaying spectra.

A. Contributions

- 1) **DSD regularization formula**—gap-adaptive damping for kernel matrix inversion, motivated by Davis-Kahan eigenvector reliability theory
- 2) **Differentiable PyTorch implementation** with gradient-optimized (α, β)
- 3) **Fair experimental comparison**—both DSD and Tikhonov receive equal optimization opportunity (50-seed paired testing)
- 4) **Clear operating regime**—DSD dominates at $d \geq 100$; equivalent at $d \leq 50$
- 5) **Honest limitations**—pre-image tasks: DSD matches but does not beat optimally-tuned Tikhonov

II. RELATED WORK

Tikhonov regularization [6] adds scalar ridge—uniform penalty regardless of spectral structure. Optimal selection of the Tikhonov parameter γ via generalized cross-validation [7] addresses the tuning problem but does not change the fundamental limitation: a single scalar applied uniformly across all eigenvectors. Truncated SVD [8] applies a hard rank cutoff. Spectral filtering [9] adapts to eigenvalue magnitude but not

eigengap structure. DSD is the only method that regularizes based on eigenvector reliability.

LSTSVM [3], [10] replaces SVM’s QP with linear systems whose matrices exhibit severe spectral tail-clustering at high dimensionality. Pre-image methods [11]–[13] require kernel matrix inversion via Nyström approximation [14], inheriting spectral instability. SHAP [15] and LIME [16] provide local model-agnostic explanations without geometric guarantees.

III. METHOD

A. DSD Formula

The DSD-regularized inverse uses eigengap-adaptive damping (Eq. 1). The full pseudo-inverse is: $\tilde{W}^+ = U \cdot \text{diag}(\tilde{\lambda}_i^{-1}) \cdot U^\top$

Behavior: Large δ_i (reliable): $\exp(-\beta\delta_i) \rightarrow 0$, exact inverse preserved. Small δ_i (corrupted): $\exp(-\beta\delta_i) \rightarrow 1$, corrupted subspace suppressed. Smooth differentiable transition between regimes.

B. Eigengap Computation

The localized eigengap uses minimum bilateral distance with single-sided boundary handling:

$$\delta_i = \begin{cases} |\lambda_1 - \lambda_2| & i = 1 \\ \min(|\lambda_i - \lambda_{i-1}|, |\lambda_i - \lambda_{i+1}|) & 1 < i < m \\ |\lambda_m - \lambda_{m-1}| & i = m \end{cases} \quad (2)$$

This ensures the smallest eigenvalue (most corrupted per Davis-Kahan) receives appropriate damping rather than being exempt due to boundary artifacts.

C. Hyperparameter Initialization

$$\alpha_0 = \lambda_{\text{transition}}^2, \quad \beta_0 = \frac{1}{\text{median}(\Delta\lambda)} \quad (3)$$

where $\lambda_{\text{transition}}$ is the eigenvalue at the 10th-percentile gap. No cross-validation required.

D. Gradient Optimization and the Structural Prior

When a differentiable reconstruction objective exists (pre-image tasks), α, β can be further refined via backpropagation. Stored in log-space as PyTorch parameters, optimized via Adam (lr=0.01, patience=20) on reconstruction loss. Typical improvement: 9–12% over principled initialization ($p < 0.0001$).

For classification (LSTSVM), the principled initialization alone provides the reported improvements *without* gradient tuning. Remarkably, optimizing α, β on reconstruction loss for LSTSVM *degrades* classification accuracy (GINA $d = 970$, 30 seeds: DSD-init 85.9% vs DSD-optimized 81.1%, $p < 10^{-6}$, init wins 30/30).

Gradient trajectory analysis reveals the mechanism: during optimization, β decreases (from 0.74 to 0.34 over 100 epochs, a $0.46\times$ reduction) while α increases ($2.3\times$). Decreasing β flattens the exponential $\exp(-\beta\delta_i)$, making damping increasingly uniform across all eigenvectors regardless of gap—converging toward constant-penalty Tikhonov behavior.

Algorithm 1 DSD-Regularized Inversion

Require: Symmetric kernel matrix $W \in \mathbb{R}^{m \times m}$

- 1: $U, \Lambda \leftarrow \text{eigh}(W)$ $\triangleright O(m^3)$
- 2: Retain $\lambda_i > 10^{-12}$ $\triangleright O(m)$
- 3: $\delta_i \leftarrow \min(|\lambda_i - \lambda_{i\pm 1}|)$ $\triangleright O(m)$
- 4: $\alpha, \beta \leftarrow \text{spectral initialization}$ $\triangleright O(m)$
- 5: $d_i \leftarrow \alpha \cdot \exp(-\beta \cdot \delta_i)$ $\triangleright O(m)$
- 6: $\tilde{\lambda}_i^{-1} \leftarrow \lambda_i / (\lambda_i^2 + d_i)$ $\triangleright O(m)$
- 7: $W^+ \leftarrow U \cdot \text{diag}(\tilde{\lambda}^{-1}) \cdot U^\top$ $\triangleright O(m^2)$

Ensure: $\tilde{W}^+ \in \mathbb{R}^{m \times m}$

The optimizer minimizes point-wise reconstruction error by removing the gap-adaptive *selectivity* that makes DSD effective for generalization. The reconstruction loss proxy is fundamentally misaligned with the classification margin: it rewards accurate inversion of *all* spectral components (including noise-corrupted ones), while classification benefits from *suppressing* those components to prevent overfitting.

This confirms DSD’s classification advantage is a **structural prior**, not a parametric advantage: the eigengap-driven initialization encodes domain knowledge about which spectral directions are trustworthy, and gradient descent on a proxy loss actively destroys this information.

E. Complexity

Total: $O(m^3)$ one-time eigendecomposition + $O(m)$ DSD computation + $O(md)$ per query. The DSD overhead is $< 0.01\%$ of total computation. For $m > 1500$, a scalable $O(m^2k)$ path using partial eigendecomposition (`scipy.sparse.linalg.eigsh`) with factored (non-materialized) inverse is provided in the implementation.

IV. EXPERIMENTAL DESIGN

A. Optimization Protocol (Per-Experiment)

For LSTSVM classification (Tables I, IV): DSD uses principled auto-initialization from the spectrum (no gradient optimization needed—the initialization alone provides the reported improvements). Tikhonov receives γ grid-searched (15-point log grid from 10^{-8} to 10^{-1}) on training accuracy.

For pre-image reconstruction (Table III): DSD receives gradient-optimized (α, β) via Adam (lr=0.01, 150 epochs). Tikhonov receives γ optimized via the *same* Adam optimizer on the same reconstruction loss (single differentiable parameter). This eliminates the grid-vs-gradient fairness concern entirely.

For Kernel-LSTSVM (Table II): DSD receives gradient-optimized (α, β) on class kernel pre-image loss. Tikhonov receives γ grid-searched (20-point log grid) on training accuracy.

All experiments: 50 seeds (30 for GINA, Madelon, and Digits due to compute cost; 20 for the gradient-Tikhonov fairness check), paired t -tests, identical data within each seed.

B. Datasets

Synthetic LSTSVM: `make_classification`: ($d = 200, n = 300$), ($d = 100, n = 200$), ($d = 50, n = 300$),

TABLE I
LSTSVM ACCURACY—DSD (AUTO-INIT) VS TIKHONOV
(GRID-SEARCHED)

Dataset	DSD	Tik-opt	Δ	Wins	C.d
GINA ($d=970$)	85.9	81.1	+4.8	30/30	4.49
Madelon ($d=500$)	57.7	55.0	+2.6	27/30	1.76
Synth. ($d=200$)	71.4	61.1	+10.4	44/50	1.57
Synth. ($d=100$)	79.1	75.8	+3.3	35/50	0.39
Synth. ($d=50$)	90.1	90.1	0	—	—

All $p < 0.0001$ where $\Delta > 0$. C.d = Cohen’s d .

Accuracy in %. Δ = advantage in pp. 30/50 seeds.

DSD: auto-init (no tuning). Tikhonov: 15-pt grid search.

($d = 30$, $n = 400$). Training noise $\sigma = 0.1$, 70/30 stratified split.

Real-World LSTSVM: GINA ($d = 970$, $n = 3468$, handwriting recognition from OpenML [10]), Madelon ($d = 500$, $n = 2600$, from OpenML—20 informative features among 500, designed to be difficult).

Pre-Image: Swiss Roll ($n = 2000$, $d = 3$), $m = 300$ landmarks, $\gamma = 2.0$, three noise levels.

Kernel-LSTSVM: Two Moons ($n = 400$, $\gamma = 2.0$), Genomics-like ($d = 100$, $n = 200$, $\gamma = 0.02$).

Boundary characterization: German Credit ($d = 20$), Breast Cancer ($d = 30$), Ionosphere ($d = 34$)—included to empirically validate the $d \leq 50$ boundary, not as DSD success claims.

V. RESULTS

A. Linear LSTSVM Classification (Primary Result)

DSD’s advantage scales with dimensionality (large effect sizes at $d \geq 200$). The GINA result yields Cohen’s $d = 4.49$ —an exceptionally large standardized effect, exceeding thresholds typically observed in regularization studies where $d > 0.8$ is considered “large” [17]. At $d \geq 100$, LSTSVM product matrices are severely ill-conditioned; the bottom 50% of eigenvalues occupy only 7–11% of the spectral range (tail-clustering), creating the regime where DSD’s per-eigenvector adaptation outperforms scalar regularization. At $d \leq 50$, tail-clustering is mild (15%+ spectral span) and all methods are equivalent.

Critically, DSD achieves these results using only its principled spectral initialization—zero optimization. Tikhonov receives a 15-point grid search (optimization advantage to the baseline). The GINA result ($d = 970$, handwriting recognition, OpenML) and Madelon result ($d = 500$, NIPS 2003 feature selection challenge) validate that DSD’s advantage transfers from synthetic to genuine high-dimensional data. Madelon’s 57.7% absolute accuracy reflects the dataset’s deliberate difficulty (20 informative among 500 features); published baselines for linear classifiers without feature selection report 55–62% on this task.

B. Why Not Per-Eigenvalue Tikhonov?

A natural question: instead of DSD’s 2-parameter formula, why not use a per-eigenvalue γ_i vector? Per-

TABLE II
KERNEL-LSTSVM—DSD-OPT VS TIKHONOV-OPT (50 SEEDS)

Dataset	DSD-opt	Tik-opt	Adv.	p
Two Moons ($\gamma=2.0$)	92.7%	92.1%	+0.6pp	0.028
Genomics ($d=100$)	87.7%	87.7%	0	0.77

TABLE III
PRE-IMAGE ERROR—DSD-OPT VS TIKHONOV-OPT (50 SEEDS, SWISS ROLL)

Noise	DSD-opt	Tik-opt	Naive	DSD vs Tik
$\sigma=5e-3$	0.069	0.069	4.56	Tie ($p=0.99$)
$\sigma=1e-3$	0.051	0.048	0.84	Tik ($p=0.02$)
$\sigma=1e-4$	0.042	0.035	0.09	Tik ($p < 0.001$)

TABLE IV
OPERATING BOUNDARY: LSTSVM AT $d \leq 64$ (DSD NOT EXPECTED TO WIN)

Dataset	d	DSD	Tik-opt	Adv.	p
<i>Below operating boundary ($d \leq 64$):</i>					
Digits	64	88.3%	89.1%	−0.9pp	Tik wins
Ionosphere	34	73.3%	80.3%	−7.0pp	Tik wins

eigenvalue Tikhonov would have m free parameters requiring m -dimensional optimization—computationally intractable. DSD’s exponential-of-eigengap structure encodes a structural prior (damping decays with spectral reliability) that reduces the search from m -dimensional to 2-dimensional while capturing the essential signal/noise boundary. The 2 parameters (α, β) implicitly define an m -dimensional damping vector through the spectral structure of the matrix itself.

C. Kernel LSTSVM (Non-Linear)

On genuinely non-linear problems with sharp RBF kernels (severe spectral decay), DSD provides statistically significant improvement even against optimally-tuned Tikhonov.

D. Pre-Image Reconstruction

For pre-image reconstruction—a task with a single scalar objective—optimally-tuned Tikhonov matches or slightly beats DSD (Table III, 50 seeds, grid-searched γ). Both methods reduce Naive error by 66× at high noise. As a secondary fairness check (20 seeds, $\sigma = 5 \times 10^{-3}$ only), we gave Tikhonov the *same* Adam gradient optimizer: DSD achieves 0.073 vs Tikhonov’s 0.078 ($p = 0.007$, DSD wins 16/20). The per-eigenvector structure provides a small but statistically significant advantage at high noise when optimization power is equalized, though the practical difference is negligible.

E. Real-World Validation and Operating Boundary

Below $d \approx 100$, DSD provides no advantage and may underperform—we include Digits ($d = 64$) and Ionosphere ($d = 34$) to empirically validate this operating boundary rather than as success claims. Practitioners should apply DSD when $d/n > 0.5$ and the system matrix condition number exceeds $\sim 10^3$.

TABLE V
WHERE DSD HELPS AND WHERE IT DOESN'T

Condition	DSD Advantage	Recommendation
LSTSVM, $d \geq 100$	+3 to +10pp	Use DSD
Kernel-LSTSVM, sharp RBF	+0.6pp	Use DSD
Pre-image (any noise)	Equivalent to Tik-opt	Either
LSTSVM, $d \leq 50$	No difference	Simplest method
Real-world, $d \leq 34$	Mixed	Test both

TABLE VI
SPECTRAL TAIL-CLUSTERING IN LSTSVM PRODUCT MATRICES

d	Cond. No.	Tail Span*	DSD Advantage
200	8.5×10^3	7.1%	+10.4pp
100	3.5×10^3	11.1%	+3.3pp
50	1.1×10^3	15.1%	0pp

*Fraction of spectral range occupied by bottom 50% of eigenvalues.

F. Operating Regime Summary

VI. ANALYSIS

A. Spectral Structure of LSTSVM Product Matrices

The spectral tail-clustering claimed for LSTSVM product matrices is verified empirically:

At $d = 200$, the bottom half of eigenvalues are compressed into 7% of the total range—extreme clustering that corrupts eigenvector directions per Davis-Kahan. At $d = 50$, the same fraction spans 15%, reducing clustering severity below DSD’s activation threshold. This directly explains the dimensionality-dependent advantage.

B. Why DSD Helps Classification But Ties on Pre-Image

The pre-image task has a single scalar objective ($\min \|\hat{x} - x\|^2$) that Tikhonov’s γ can optimize directly via grid search. Classification has a more complex objective (generalization to unseen data) where per-eigenvector regularization prevents the model from fitting to noise-corrupted spectral directions. This spectral regularization effect—suppressing unreliable directions during *training*—cannot be replicated by a single scalar applied uniformly.

At $d = 200$, LSTSVM product matrices have ~ 200 eigenvalues with the bottom $\sim 50\%$ tightly clustered. DSD suppresses these (preventing noise-fitting) while Tikhonov’s single γ either under-regularizes the tail or over-regularizes the reliable head. This trade-off is impossible to resolve with one parameter.

C. Connection to Generalization Theory

DSD’s classification advantage can be understood through the lens of algorithmic stability. In LSTSVM, the solution $u = M^{-1} \cdot \text{rhs}$ depends on all eigenvectors of M . Under Tikhonov, a single training perturbation propagates through *all* spectral directions with uniform damping γ , including those with small gaps where Davis-Kahan guarantees large directional sensitivity. DSD suppresses contributions from these sensitive directions, effectively reducing the hypothesis class

to the subspace spanned by well-separated eigenvectors. This spectral truncation (soft, not hard) reduces the effective dimensionality of the learner: fewer spectral directions participate in the solution, bounding the model’s sensitivity to individual training samples—the hallmark of algorithmic stability that implies generalization [18]. A formal Rademacher complexity bound incorporating DSD’s gap-dependent damping profile remains an open theoretical direction.

D. Computational Overhead and Memory

DSD adds $O(m)$ computation to the existing $O(m^3)$ eigendecomposition—less than 0.01% overhead. Total cost is identical to Tikhonov within measurement precision.

For the differentiable PyTorch implementation, the dominant memory cost is the backward pass through `torch.linalg.eigh`, which requires storing the full eigenvector matrix $U \in \mathbb{R}^{m \times m}$ and its gradient. Peak GPU memory during backpropagation is approximately $4m^2 \times 8$ bytes (float64): ~ 31 MB at $m = 1000$ and ~ 763 MB at $m = 5000$. For inference-only deployment (no gradient), memory is $2m^2 \times 8$ bytes (W and $\tilde{W}^+$). The scalable factored path (`DSDScalableResult.apply()`) avoids materializing $\tilde{W}^+$ entirely, requiring only $O(mk)$ storage where $k \ll m$.

VII. LIMITATIONS AND FUTURE WORK

A. Limitations

- 1) **No formal stability proof**—DSD boundedness as $\delta_i \rightarrow 0$ remains conjectured
- 2) **Non-RBF kernels**—initialization assumes monotonic decay; not evaluated on polynomial or non-stationary kernels
- 3) **Pre-image not superior**—matches but does not beat optimally-tuned Tikhonov
- 4) **Real-world results mixed**—at $d \leq 34$, advantage is dataset-dependent
- 5) **Eigendecomposition gradient instability**—PyTorch `eigh` backward produces ill-conditioned gradients for degenerate eigenvalues; mitigated via gradient clipping

B. Future Work

Three directions follow naturally. First, a **Tikhonov-DSD hybrid** that applies Tikhonov to well-separated eigenvalues and DSD to the clustered tail would eliminate the regime where Tikhonov slightly wins (low noise, low dimension). Second, **other cross-class product matrix methods**—kernel Fisher discriminant analysis, kernel CCA, and structured output SVMs—share LSTSVM’s property of constructing system matrices from cross-class interactions with severe spectral tail-clustering; DSD’s advantage should transfer directly to these. Third, a formal **stability proof** combining Davis-Kahan perturbation bounds with DSD’s damping structure to show that pre-image error remains bounded regardless of spectral clustering would complete the theoretical foundation.

VIII. CONCLUSION

DSD is a spectral-structure-aware regularizer for ill-conditioned kernel methods. Its primary contribution is **classification accuracy on high-dimensional LSTSVM problems**: +4.8pp on GINA ($d = 970$, Cohen’s $d = 4.49$), +10.4pp at $d = 200$, and +2.6pp on Madelon ($d = 500$)—all over optimally-tuned Tikhonov with $p < 0.0001$, using only principled initialization (no optimization). The gap-adaptive damping prevents overfitting to noise-corrupted eigenvector directions during training—an effect that scalar regularization cannot achieve. Gradient optimization on reconstruction loss actively undoes this beneficial regularization (GINA: 85.9% init vs 81.1% optimized), confirming the advantage is structural rather than parametric.

DSD does not beat Tikhonov on pre-image reconstruction (they are equivalent), and does not help at $d \leq 50$ where spectral tail-clustering is insufficient. It is a tool for the high-dimensional, ill-conditioned regime—which is precisely where LSTSVM is most needed (small-sample genomics, spectroscopy, high-dimensional feature engineering).

The method is computationally free ($O(m)$ overhead), requires no cross-validation, and applies to any kernel method whose system matrices exhibit spectral tail-clustering.

Reproducibility

Code, all experiment scripts, and raw results are publicly available at <https://github.com/Praveg432/dsd-regularization>.

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